

23AHS21 - probability and statistics

Assignment-1

1) Define i) statistics ii) population

iii) Sample iv) primary data v) Secondary data
Collection. with an example. vi) Range.

2) Find Mean, Median, mode, Geometric Mean and Harmonic Mean from the following data:

profits per shop (Rs.)	100-200	200-300	300-400	400-500	500-600	600-700	700-800
Number of Shops.	10	18	20	26	30	28	18

3) Calculate the variance, Standard deviation and Mean deviation about the mean for the following data.

class interval	30-40	40-50	50-60	60-70	70-80	80-90	90-100
Frequency.	3	7	12	15	8	3	2

4) From the following distribution, Calculate
i) First four moments about the mean
ii) Skewness based on moments.
iii) Kurtosis.

Income (Rs)	0-10	10-20	20-30	30-40
Frequency	1	3	4	2

5) i) Calculate Karl Pearson's Correlation Coefficient for the following data.

X	65	66	67	67	68	69	70	72
Y	67	68	65	68	72	72	69	71

ii) A random sample of five College students is selected and their grades in Mathematics and Statistics are found to be

	1	2	3	4	5
Mathematics	85	60	73	40	90
Statistics	93	75	65	50	80

Calculate Spearman's Rank Correlation Coefficient.

iii) For the following data, find equations of the two regression lines.

X	1	2	3	4	5
Y	15	25	35	45	55